

Restless nights: gender, age, and happiness as determinants of sleep quality in France and Sweden

Sleep is a fundamental aspect of human health, yet restless sleep has become a widespread concern across Europe. Sociological research demonstrates that modern rest is deeply shaped by socio-demographic structures and psychological states, rather than being a purely biological phenomenon.

Academic literature consistently highlights a significant gender gap, showing that women report higher levels of restless sleep than men, due to the unequal pressure of social roles and daily stressors (Groeger et al., 2004; Baranowski & Jabkowski, 2023). Furthermore, age is heavily associated with sleep degradation, making older individuals particularly vulnerable to fragmented nights (Groeger et al., 2004). Beyond demographic traits, subjective well-being operates as a critical determinant of rest. High life satisfaction acts as a psychological shield, whereas lower levels of happiness induce cognitive arousal and stress, preventing nocturnal relaxation (Ness & Saksvik-Lehouillier, 2018). Crucially, when these vulnerabilities intersect, specifically among older women with lower happiness, the cumulative impact of ageing, social roles, and psychological distress is expected to generate the highest risk of severe sleep disruption. Finally, these dynamics are likely to vary across national contexts due to differing institutional and welfare frameworks between France and Sweden (Baranowski & Jabkowski, 2023).

Based on these premises, this essay tests the following hypotheses:

- H1: Women report more restless sleep than men.
- H2: Older individuals report higher levels of restless sleep.
- H3: Levels of restless sleep differ between France and Sweden.
- H4: Older women with lower levels of happiness report the highest levels of restless sleep.

For this analysis, we rely on data collected by ESS: CRONOS-2 Wave 4 edition 1.0. The analytical sample is restricted to adult respondents residing in France and Sweden, yielding a final sample of 1,525 observations (France = 705, Sweden = 820) after listwise deletion of missing values on all variables of interest.

The dependent variable is restless sleep (firstly named *w4q49*, then renamed *rstless*), measuring how often respondents experienced disrupted rest during the past week, scored on a four-point ordinal scale ranging from 1 ("Rarely or none of the time") to 4 ("Most or all of the time"). The primary socio-demographic independent variables are gender, captured as a dummy variable (0 = men, 1 = women), and age, treated as a continuous variable. National context is accounted for using a country dummy variable (0 = France, 1 = Sweden). Finally, subjective well-being is measured through self-reported happiness, also scored on a four-point ordinal scale ranging from 1 ("Rarely or none of the time") to 4 ("Most or all of the time").

The analysis plan consists of two steps. First, to test the gender gap hypothesis (H1), we run a two-sample t-test comparing mean levels of restless sleep between women and men. Second, to test

hypotheses H2, H3, and the intersection captured in H4, we estimate linear regression models. The baseline model includes age, country, and gender as predictors. The final model adds a three-way interaction term between gender, age, and happiness (firstly named *w4q50*, then renamed as *happiness*) to formally evaluate whether older, less happy women report the highest levels of restless sleep.

Results

The analytical sample consists of 1,525 respondents, 705 from France and 820 from Sweden, with women representing approximately 53% of the total. The average age is around 51 years, reflecting a relatively mature sample (Table 1).

Regarding the key variables, restless sleep shows a mean of approximately 2 on the four-point scale, suggesting that respondents experience disrupted sleep "some of the time" on average. Happiness scores are higher, with a mean around 2.7, indicating that respondents tend to feel happy "often" (Table 1).

A first look at the data already offers some preliminary evidence in line with our hypotheses.

- Sweden shows marginally lower levels compared to France: 1.81 vs 2.20 (Table 2a).
- Women report slightly higher levels of restless sleep than men: 2.07 vs 1.91 (Table 2b).
- Younger respondents (18-34) tend to report slightly higher levels of restless sleep than older ones (55+): 2.15 vs 1.83. (Table 2c).

These patterns are not conclusive on their own, but provide a useful baseline before moving to the inferential analyses.

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. * Overall summary
. summarize rstless happiness agea female sweden
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Variable	Obs	Mean	Std. dev.	Min	Max
rstless	1,525	1.992131	.894686	1	4
happiness	1,525	2.655738	.8060883	1	4
agea	1,525	51.10426	16.67684	18	90
female	1,525	.5259016	.4994924	0	1
sweden	1,525	.5377049	.4987399	0	1

Table 1. Overall summary

<pre>. * By country . tabstat rstless happiness agea female, by(sweden) stat(mean sd n)</pre> Summary statistics: Mean, SD, N Group variable: sweden (Country (1=Sweden)) <table><tr><th>sweden</th><th>rstless</th><th>happin-s</th><th>agea</th><th>female</th></tr><tr><td>France</td><td>2.198582</td><td>2.646809</td><td>47.95603</td><td>.5361702</td></tr><tr><td></td><td>.8824348</td><td>.7491049</td><td>15.69143</td><td>.4990441</td></tr><tr><td></td><td>705</td><td>705</td><td>705</td><td>705</td></tr><tr><td>Sweden</td><td>1.814634</td><td>2.663415</td><td>53.81098</td><td>.5170732</td></tr><tr><td></td><td>.8671346</td><td>.8524206</td><td>17.02959</td><td>.5000134</td></tr><tr><td></td><td>820</td><td>820</td><td>820</td><td>820</td></tr><tr><td>Total</td><td>1.992131</td><td>2.655738</td><td>51.10426</td><td>.5259016</td></tr><tr><td></td><td>.894686</td><td>.8060883</td><td>16.67684</td><td>.4994924</td></tr><tr><td></td><td>1525</td><td>1525</td><td>1525</td><td>1525</td></tr></table>					sweden	rstless	happin-s	agea	female	France	2.198582	2.646809	47.95603	.5361702		.8824348	.7491049	15.69143	.4990441		705	705	705	705	Sweden	1.814634	2.663415	53.81098	.5170732		.8671346	.8524206	17.02959	.5000134		820	820	820	820	Total	1.992131	2.655738	51.10426	.5259016		.894686	.8060883	16.67684	.4994924		1525	1525	1525	1525	<pre>. * By gender . tabstat rstless, by(female) stat(mean sd n)</pre> Summary for variables: rstless Group variable: female (Gender (1=Female)) <table><tr><th>female</th><th>Mean</th><th>SD</th><th>N</th></tr><tr><td>Male</td><td>1.905947</td><td>.8596885</td><td>723</td></tr><tr><td>Female</td><td>2.069825</td><td>.9187258</td><td>802</td></tr><tr><td>Total</td><td>1.992131</td><td>.894686</td><td>1525</td></tr></table>					female	Mean	SD	N	Male	1.905947	.8596885	723	Female	2.069825	.9187258	802	Total	1.992131	.894686	1525	<pre>. * By age group . tabstat rstless, by(agegroup) stat(mean sd n)</pre> Summary for variables: rstless Group variable: agegroup (Age group) <table><tr><th>agegroup</th><th>Mean</th><th>SD</th><th>N</th></tr><tr><td>Young (18-34)</td><td>2.153846</td><td>.9610498</td><td>286</td></tr><tr><td>Middle (35-54)</td><td>2.103147</td><td>.9062097</td><td>572</td></tr><tr><td>Older (55+)</td><td>1.827586</td><td>.8267055</td><td>667</td></tr><tr><td>Total</td><td>1.992131</td><td>.894686</td><td>1525</td></tr></table>					agegroup	Mean	SD	N	Young (18-34)	2.153846	.9610498	286	Middle (35-54)	2.103147	.9062097	572	Older (55+)	1.827586	.8267055	667	Total	1.992131	.894686	1525
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Table 2. Results by country (2a), gender (2b), and age (2c)

T-test - H1: gender differences in restless sleep

The two-sample t-test compares mean levels of restless sleep between women and men across the pooled sample of the two countries. The results provide clear support for H1: women report

significantly higher levels of restless sleep than men. The difference, while modest in absolute terms, is statistically significant, meaning it is very unlikely to be due to chance alone.

When looking at the two countries separately, the gender gap holds in both France and Sweden, confirming that this is not a country-specific pattern but rather a consistent finding across the two national contexts. In both cases, the confidence interval does not contain zero and the p-value falls well below the conventional 0.05 threshold, allowing us to reject the null hypothesis of no difference between genders (Table 3).

These results confirm the unequal distribution of social roles and daily stressors as key drivers of women's poorer sleep quality (Figure 1).

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. * H1: Women report more restless sleep than men
. * Two-sample t-test: sleep by gender (pooled FR + SE)
. ttest rstless, by(female)

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Group	Obs	Mean	Std. err.	Std. dev.	[95% conf. interval]	
Male	723	1.905947	.0319722	.8596885	1.843178	1.968717
Female	802	2.069825	.0324413	.9187258	2.006145	2.133506
Combined	1,525	1.992131	.0229106	.894686	1.947192	2.037071
diff		-.163878	.0457053		-.25353	-.074226

diff = mean(Male) - mean(Female)		t = -3.5855
H0: diff = 0		Degrees of freedom = 1523
Ha: diff < 0	Ha: diff != 0	Ha: diff > 0
Pr(T < t) = 0.0002	Pr(T > t) = 0.0003	Pr(T > t) = 0.9998

Table 3. Two-sample t-test for H1

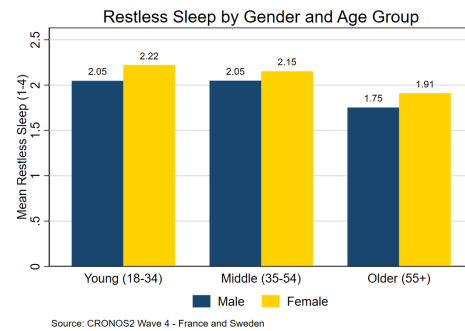


Figure 1. Graph of restless sleep by gender and age group

Linear regression - H2 and H3: age and country effects

The baseline regression model includes age, country, and gender as predictors of restless sleep. Starting with H2, the coefficient on age is negative and statistically significant ($\beta = -0.007$, $p < 0.001$), indicating that restless sleep actually decreases with age in this sample. This result does not support H2 and contradicts the expected direction. This unexpected finding may reflect compositional features of the CRONOS2 panel and does not allow us to confirm H2 (Table 4).

Turning to H3, the coefficient on the Sweden dummy is negative and statistically significant ($\beta = -0.343$, $p < 0.001$), indicating that Swedish respondents report lower levels of restless sleep compared to their French counterparts, even after controlling for age and gender. This difference likely reflects the broader institutional and welfare context of the two countries, with Sweden's stronger social protection system potentially contributing to lower stress levels and better sleep outcomes (Table 4). Finally, the coefficient on gender remains positive and significant ($\beta = 0.133$, $p = 0.003$) even after controlling for age and country, further reinforcing the evidence in favour of H1 found in the t-test (Table 3; Table 4).

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. * 6. H2 + H3: Age and country effects (linear regression) *
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. * H2: Older people report more restless sleep
. * H3: Sleep differs between France and Sweden
. * Baseline OLS: restless sleep ~ age + country + gender
.
. regress rstless agea sweden female

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Source	SS	df	MS	Number of obs	=	1,525
Model	82.8934101	3	27.6311367	F(3, 1521)	=	36.96
Residual	1137.01216	1,521	.747542514	Prob > F	=	0.0000
				R-squared	=	0.0680
				Adj R-squared	=	0.0661
Total	1219.90557	1,524	.800462975	Root MSE	=	.86461

rstless	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
agea	-.0066003	.0013569	-4.86	0.000	-.009262	-.0039386
sweden	-.3427627	.0451038	-7.60	0.000	-.4312349	-.2542904
female	.1330215	.0446128	2.98	0.003	.0455123	.2205307
_cons	2.443784	.0787758	31.02	0.000	2.289263	2.598304

Table 4. Linear regression for H2-H3

Linear regression with three-way interaction - H4: gender, age, and happiness

The final model includes a three-way interaction between gender, age, and happiness, while controlling for country. This is the most complex part of the analysis and directly tests H4, namely, whether older women with lower happiness report the highest levels of restless sleep.

Interpreting a three-way interaction requires looking at the combined effect of all three variables simultaneously rather than each coefficient in isolation. The interaction term between gender, age, and happiness captures whether the joint effect of being older and less happy on restless sleep differs between women and men.

The results show that the three-way interaction term is not statistically significant ($\beta = -0.004$, $p = 0.156$), meaning that H4 cannot be formally confirmed. However, the direction of the coefficient is consistent with the expected pattern: the negative sign suggests that, among women, being older and less happy is associated with more restless sleep, even if this difference does not reach the conventional 0.05 significance threshold. In other words, older women who rarely feel happy appear to be the most vulnerable group in terms of sleep disruption, which is consistent with the cumulative disadvantage argument outlined in the introduction. This may reflect the complexity of the interaction rather than the absence of the phenomenon altogether.

The country control remains significant also in this model ($\beta = -0.339$, $p < 0.001$), confirming that the France-Sweden difference in restless sleep persists even after accounting for the complex interplay of gender, age, and happiness (Table 5).

Overall, the evidence does not allow us to formally support H4, though the observed pattern points in the hypothesised direction.

	rstless	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
female							
Female		-.6311577	.4747168	-1.33	0.184	-1.562329	.3000136
agea		-.0059127	.0066056	-0.90	0.371	-.0188696	.0070443
female#c.agea							
Female		.0139983	.0087669	1.60	0.111	-.0031983	.0311949
happiness		-.3122152	.1281644	-2.44	0.015	-.5636135	-.0608169
female#c.happiness							
Female		.2426327	.1686544	1.44	0.150	-.088188	.5734533
c.agea#c.happiness		-.0006881	.0023216	-0.30	0.767	-.0052421	.0038658
female#c.agea#c.happiness							
Female		-.0044179	.0031094	-1.42	0.156	-.0105171	.0016814
sweden		-.3390099	.0428632	-7.91	0.000	-.4230874	-.2549324
_cons		3.333728	.3647462	9.14	0.000	2.618268	4.049189

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. * 7. H4: Three-way interaction gender x age x happiness *
. *-----*
.
. * H4: The effect of age on restless sleep differs by gender
. * and is moderated by happiness
.
. * Full model with three-way interaction, controlling for country
. regress rstless i.female#c.agea#c.happiness sweden

```

Source	SS	df	MS	Number of obs	=	1,525
				F(8, 1516)	=	37.05
Model	199.487281	8	24.9359102	Prob > F	=	0.0000
Residual	1020.41829	1,516	.673099137	R-squared	=	0.1635
				Adj R-squared	=	0.1591
Total	1219.90557	1,524	.800462975	Root MSE	=	.82043

Table 5. Linear regression for H4

Conclusions

This short paper looked at gender, age, and happiness as factors linked to restless sleep in France and Sweden, using CRONOS2 Wave 4 data.

The t-test results clearly show a difference in restless sleep between women and men, both overall and in each country separately. This supports H1 and confirms that unequal social roles and daily stress are likely behind women's poorer sleep quality.

H2, however, is not supported: restless sleep actually decreases with age instead of increasing, which goes against what was expected. This may be due to specific features of the CRONOS2 sample rather than a real absence of an age effect.

H3 is supported instead, as Swedish respondents report lower restless sleep than French respondents, even after accounting for age and gender. This is likely linked to Sweden's stronger welfare system compared to France.

Ultimately, H4 is not confirmed: the three-way interaction between gender, age, and happiness is not statistically significant, although it points in the expected direction, suggesting that older women with lower happiness may still be more vulnerable to restless sleep.

These results suggest that future research, ideally with larger and longer-term samples, is needed to better understand how age and well-being affect sleep quality.

References

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Groeger, J. A., Zijlstra, F. R., & Dijk, D. J. (2004). Sleep quantity, quality, and working sample characteristics. *Journal of Sleep Research*, 13(4), 359-363.

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